

SMART INDIA HACKATHON 2026



Water Intelligence Project Guide

Beginner Roadmap & Execution Playbook for Saurav & Akansha

1 What is Our Project in Simple Words?

Water Intelligence is a smart web platform that predicts water shortages **before** they happen. Instead of just showing today's water numbers, it looks at **rainfall drops + water usage increases + reservoir levels** and gives early warnings (Low, Moderate, High, Critical) so authorities can take action on time.

2 Your Individual Roles & Responsibilities

FRONTEND & VISUALISATION

Saurav

"I build what the judges and users see."

- ✓ **The Dashboard UI:** Location dropdown, KPI cards, and risk alert badges.
- ✓ **Data Visualisations:** 4-day forecast bar chart & 5-week historical line charts using **Chart.js**.
- ✓ **Interactive Map:** Visual map of India with colored risk pins using **Leaflet.js**.
- ✓ **Connecting to Backend:** Using JavaScript `fetch()` to load data from Akansha & Deep's API.

DATABASE & DATA MANAGEMENT

Akansha

"I structure and store the data safely."

- ✓ **Database Schema:** Designing tables for Regions, Daily Rainfall, Water Storage, and Alerts using **SQLite / SQL**.
- ✓ **Data Organisation:** Storing historical datasets and sensor feeds in an organized structure.
- ✓ **Data Queries:** Writing SQL queries (`SELECT`, `INSERT`, `FILTER` by date/region).
- ✓ **Supplying Data:** Serving clean records to Deep's ML model and Saurav's frontend API.

3 How Saurav & Akansha Connect Together

The 3-Step Handshake:

1. **Akansha** sets up the SQLite database containing tables for water levels, rainfall, and districts.
2. **Deep & Bhavishya** read data from Akansha's database, run AI predictions, and return a JSON output.
3. **Saurav** takes that JSON in JavaScript and renders the high-tech charts and alert banners on screen!

4 Akansha's Core Database Tables (Simplified SQLite)

```
-- Table 1: Regions & Telemetry CREATE TABLE regions ( region_id TEXT PRIMARY KEY, -- e.g. 'delhi', 'jaipur' region_name TEXT, -- 'Delhi NCR' state TEXT, -- 'Delhi' current_storage_pct REAL, -- 34.0 groundwater_depth_m REAL, -- 38.5 risk_level TEXT -- 'LOW', 'MODERATE', 'HIGH', 'CRITICAL' ); -- Table 2: 4-Day Forecast Output (Saurav displays this in Chart.js) CREATE TABLE forecast_4days ( id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
region_id TEXT, day_label TEXT, -- 'Today', 'Day 2', 'Day 3', 'Day 4' availability_pct REAL, -- 58, 48, 38,
28 status_badge TEXT -- 'Moderate', 'Stressed', 'High Risk', 'Critical' );
```

5 Step-by-Step 7-Day Study Plan for Beginners

Timeline	Saurav (Frontend & Visualisation)	Akansha (Database & Management)
Days 1 - 2	HTML & CSS Basics: Learn <code><select></code> , <code><canvas></code> , <code><div></code> , and basic Tailwind CSS classes for cards and colors.	SQL Fundamentals: Learn basic SQL syntax: <code>CREATE TABLE</code> , <code>INSERT INTO</code> , <code>SELECT * FROM</code> , and <code>WHERE</code> clauses.
Days 3 - 4	JavaScript Basics: Learn how to change text on a page using <code>document.getElementById()</code> and listen for clicks.	SQLite & DB Browser: Download <i>DB Browser for SQLite</i> . Practice creating the regions and forecast tables manually.
Days 5 - 6	Chart.js & Leaflet: Learn how Chart.js takes an array of numbers (e.g. <code>[58, 48, 38, 28]</code>) and turns it into colored bars.	Python + SQLite: Learn how to connect Python to SQLite using <code>import sqlite3</code> and fetch rows as JSON dictionary objects.
Day 7	Connect with Team: Test switching regions in the dashboard and test the "Simulate AI Shift" button for demo day.	Data Pipeline Integration: Ensure the database can easily give data to Deep's Flask backend to pass to Saurav.

6 What to Say to the SIH Judges (Your 30-Second Answers)

SAURAV'S 30-SECOND JUDGE ANSWER

"Hello Judges! I built the **Frontend and Data Visualisation layer** of Water Intelligence. When a user or authority selects a region, my frontend dynamically renders the real-time risk level, the 4-day predictive availability bar chart using Chart.js, the geographic risk map using Leaflet.js, and actionable early-warning alerts. My interface makes complex environmental data simple and actionable for decision-makers."

AKANSHA'S 30-SECOND JUDGE ANSWER

"Hello Judges! I am responsible for the **Database Architecture and Data Management layer**. I designed the structured schema in SQLite to store regional telemetry, rainfall anomaly feeds, reservoir storage metrics, and the multi-day AI prediction logs. My module ensures that data is cleanly organized, indexed for fast querying, and seamlessly served to our AI pipeline and user dashboard."

7 🏆 Golden Tips for SIH 2026 Success

1. Confidence & Clarity:

Judges care more about understanding the flow of data than how complicated your code is. Keep your explanation crisp!

2. Keep Fallback Data Ready:

During hackathons, internet or local servers can glitch. Having your mock database and dashboard ready ensures 0% demo failure.